

Streamflow reconstruction in Southern Italy: Hydrological Models versus Machine Learning Approaches

Giacomo Nicoletti^{1*}, Antonia Longobardi¹, Evangelos Bagkis², Theodosios Kassandros² and Kostas Karatzas²

¹ Department of Civil Engineering, University of Salerno, Via Giovanni Paolo II, 132, 84084 Fisciano, Italy. *E-mail: ginicoletti@unisa.it

² Environmental Informatics Research Group, School of Mechanical Engineering, Aristotle University of Thessaloniki, 54124 Thessaloniki, Greece

INTRODUCTION

- Continuous long-term hydrological records are essential for understanding hydro-climatic processes and supporting sustainable water resources management. However, monitoring networks frequently suffer from **spatial and temporal gaps**, an issue that is particularly critical for **observed streamflow data**, which cannot be directly measured but is derived through rating curves requiring periodic updating over time.
- In **data-scarce contexts**, incorporating information from **neighbouring or hydrologically similar basins** can represent a valuable modelling strategy.



Research Question: can a data-driven regionalization approach represent a viable alternative to locally calibrated hydrological modelling for streamflow reconstruction?

STUDY AREA AND DATA

- Study Area:** The Campania region (Southern Italy) underwent a structural reconstruction of its hydroclimatic monitoring network in the early 2000s, after which observed streamflow data became unavailable. Based on historical station consistency, four major catchments were selected for the analysis: Sele, Tanagro, Volturno, and Calore (Fig. 1, Tab. 1).
- Data and Temporal Window:** historical records were analysed to identify a common temporal window of concurrent observations. The selected period was 1965 – 1970, ensuring complete precipitation, temperature, and discharge data availability across all four catchments.



Basin	ID	Area (km ²)	Channel pattern
Sele	NA27	3216	Meandering
Tanagro	NA25	660	Sinuuous
Volturno	NA21	5586	Meandering
Calore	NA20	2991	Sinuuous

Figure 1 (left) and Table 1 (top). Basin's geography and characteristics.

PRE-PROCESSING

- Self-Organizing Map (SOM):** applied using variance-normalised daily precipitation, temperature, and streamflow data, with streamflow further log-transformed and converted to mm/d.
- K-means Clustering:** applied to the SOM output to evaluate inter-catchment relationships and identify hydroclimatic regimes across the four basins.
- SOM and K-means clustering **identified** the main **hydroclimatic regimes**, distinguishing high-flow and low-flow conditions, and revealed both similarities and differences among catchments in terms of **climatic characteristics** and **hydrological response**.

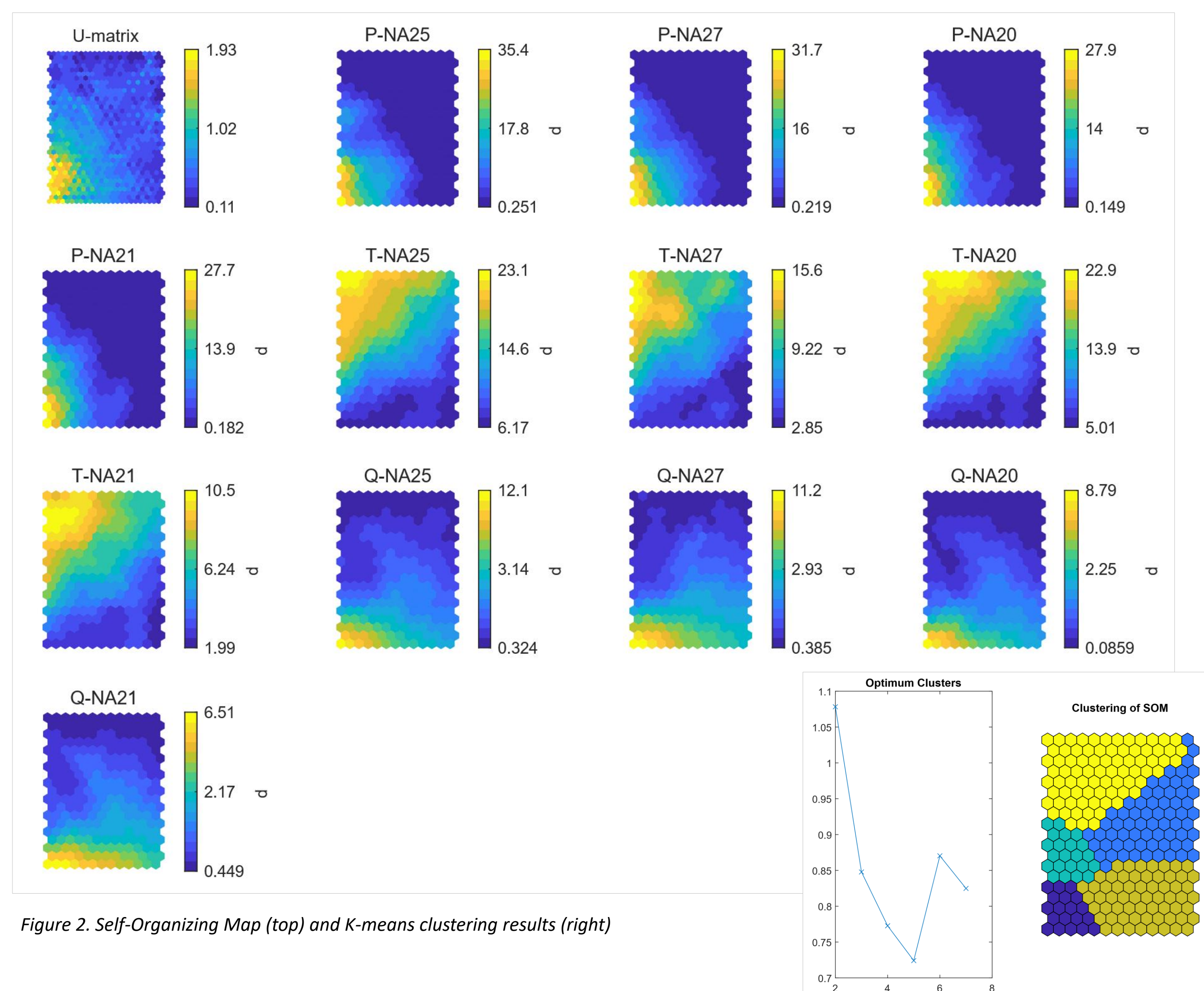


Figure 2. Self-Organizing Map (top) and K-means clustering results (right)

MODELLING

Rainfall-runoff models:

- TOPMODEL: requires daily precipitation and potential evapotranspiration as input and involves the calibration of 7 parameters.
- IHACRES: uses daily precipitation and temperature and requires the calibration of 6 parameters

Machine learning approaches:

- Random Forest Regressor: an ensemble learning method that combines multiple decision trees to improve predictive accuracy and robustness.
- XGBoost: a gradient boosting algorithm that builds an ensemble of decision trees sequentially, each correcting the residuals of the previous.
- Input for both ML models: daily precipitation, temperature and observed discharge, consistent with the data used in hydrological modelling.

Model Fitting and Validation:

- The first three years (1965–1968) served as the calibration period for TOPMODEL and IHACRES, each applied at the individual catchment scale, and as the training period for the ML models; the following two years (1969–1970) were used for validation. Model performance was assessed using NSE, RMSE, and MAE.
- ML models were evaluated in a regionalization framework in which each basin was in turn held out as the target, with the remaining three serving as donors for training.

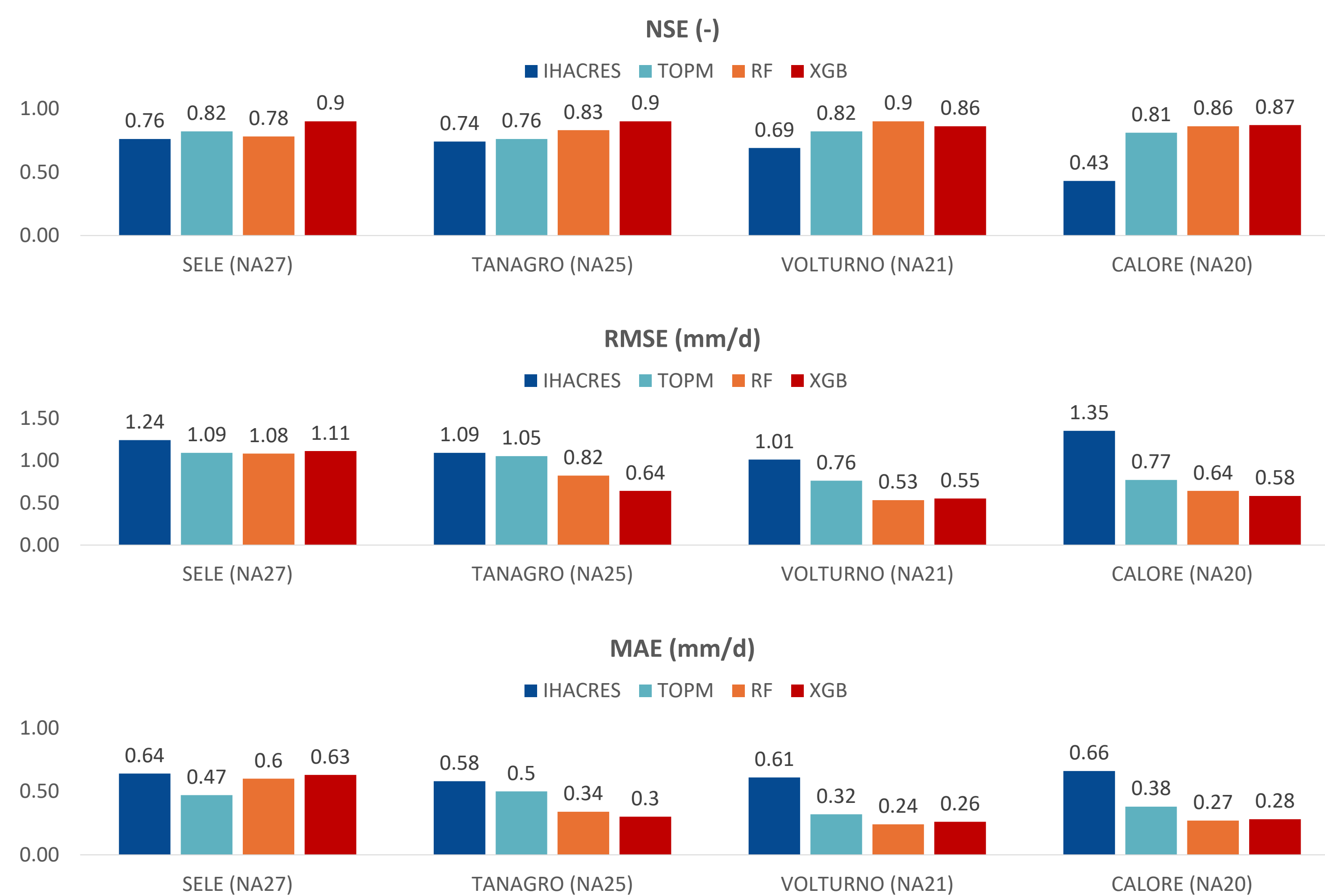


Figure 3. Performance metrics of basin-scale calibrated HL models and ML regional approach across the four catchments.

CONCLUSION AND FUTURE STEPS



Machine Learning models showed improved performance compared to hydrological modelling, suggesting that using information from donor basins improves the accuracy of streamflow reconstruction.

- The regionalization framework requires a common overlapping observation period across basins, which may limit its applicability in **highly data-scarce contexts**.
- Future steps include a **sensitivity analysis** of model outputs to individual input variables, to better characterize predictor contributions, and the integration of **water level records** as additional inputs, generally available over longer historical periods than discharge, further **extending the potential of the approach**.

References

- Breiman L (2001) Random forests. Machine Learn 45(1):5–32
- Chen T, Guestrin C (2016) XGBoost: a scalable tree boosting system. Proc 22nd ACM SIGKDD Int Conf Knowl Discov Data Min, pp 785–794
- Jakeman AJ, Hornberger GM (1993) How much complexity is warranted in a rainfall-runoff model? Water Resour Res 29:2637–2649
- Perrin C (2002) Vers une amélioration d'un modèle global pluie-débit au travers d'une approche comparative. La Houille Blanche 88:84–91
- Vesanto, J., Himberg, J., & Alhoniemi, E. (2000). SOM Toolbox for Matlab 5. (Helsinki University of Technology, Publications in Computer and Information Science; No. Report A57).